

An Internship Report
Of
Data Science Master Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410031047

Name of Student: ANJALI KUMARI

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, **ANJALI KUMARI**, hereby declare that this internship report titled "Data Science Master Virtual Internship" has been prepared as a record of my participation in the eight-week virtual internship program. It summarises the learning areas, practical activities, tools and assessment followed during the internship.

The internship was completed during April–June 2026 under the AICTE–EduSkills Virtual Internship Program in the Data Science Master domain. The supplied certificate confirms successful completion of the program.

Signature:-

Student Name: ANJALI KUMARI

Roll No.: 2410031047

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the organisations and academic institutions associated with the Data Science Master Virtual Internship for providing a structured opportunity to develop practical knowledge in data science.

I am thankful to the AICTE–EduSkills Virtual Internship Program and EduSkills Academy for providing the learning platform, study modules and assessment framework. I also acknowledge Siemens Industry Software (India) Pvt. Ltd. for supporting the program.

I am grateful to IILM University, Greater Noida, and the School of Computer Science and Engineering for their academic support and encouragement. The internship helped me relate classroom concepts to a practical data science workflow.

Signature:-

Student Name: ANJALI KUMARI

Roll No.: 2410031047

TABLE OF CONTENTS

SL. No	Section	Page no.
1	Candidate's Declaration	i
2	Acknowledgement	ii
3	Internship Completion Certificate	-
4	Project Description	1
4.1	Introduction	1
4.2	Program Profile	1
4.3	Problem Statement	1
4.4	Project Objectives	2
4.5	Scope of the Project	2
4.6	Technologies and Tools Used	2
4.7	System Architecture	3
4.8	Methodology	3
4.9	Expected Outcomes	4
4.10	Certificate of Completion	4
5	Bibliography/References	5

3. INTERNSHIP COMPLETION CERTIFICATE



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



भारत सरकार
GOVERNMENT OF INDIA
NATIONAL
INTERNSHIP
PORTAL



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Anjali Kumari

IILM University, Greater Noida

has successfully completed the 8-weeks

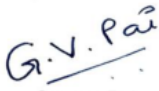
Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

Supported By

SIEMENS



Gaurav Pai
Head - Strategy and Operations
Siemens Industry Software (India) Pvt. Ltd



Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education



Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 420d992094f9d74a558e
Student ID : anjali.kumari.cs28@iilm.edu



GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the learning experience completed through the Data Science Master Virtual Internship during April–June 2026. The program introduced a practical data science workflow, beginning with Python and data preparation and progressing towards visualisation, statistics and machine learning.

The internship used a virtual learning environment combining learning modules, practical activities, quizzes and a final assessment. The sequence helped connect individual technical concepts into a complete data-oriented process.

4.2 Program Profile

The internship was conducted under the AICTE–EduSkills Virtual Internship Program in the Data Science Master domain. The supplied certificate states that Anjali Kumari successfully completed the eight-week internship during April–June 2026 and received an Outstanding grade.

The certificate identifies EduSkills and the AICTE National Internship Portal and shows support from Siemens Industry Software (India) Pvt. Ltd. The academic institution mentioned on the certificate is IILM University, Greater Noida.

4.3 Problem Statement

The learning problem addressed by the internship was the difficulty of applying separate programming, statistics and machine learning concepts as one connected workflow. The internship provided a structured path from data collection and preparation to analysis, model building and evaluation.

4.4 Project Objectives

- Strengthen Python programming for data analysis.
- Practise data cleaning and manipulation with NumPy and Pandas.
- Understand visualisation techniques using Matplotlib and Seaborn.
- Build familiarity with probability and statistical concepts.
- Explore supervised and unsupervised machine learning methods.
- Practise feature engineering and model evaluation.
- Combine the learning areas in a capstone-style workflow.
- Complete the final internship assessment and credential validation.

4.5 Scope of the Project

The internship covered the major stages of a data science workflow, including Python programming, data preparation, exploratory analysis, visualisation, statistics, machine learning and model evaluation. Activities were performed in the virtual learning environment provided by the program and were not live production deployment.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Programming and data-analysis tasks
NumPy	Numerical operations and array processing
Pandas	Data loading, cleaning and manipulation
Matplotlib	Charts and graphical analysis
Seaborn	Statistical visualisation
scikit-learn	Machine learning and model evaluation
Jupyter / Google Colab	Notebook-based development and practice
Git / GitHub	Version control and collaboration

4.7 System Architecture

The internship workflow can be organised into connected layers, with each stage preparing the learner for the next stage:

- **Data Collection Layer:** gathering datasets and understanding their structure.
- **Data Preparation Layer:** cleaning, transforming and organising data.
- **Exploratory Analysis Layer:** using summaries and visualisations to identify patterns.
- **Modelling Layer:** applying supervised and unsupervised machine learning methods.
- **Evaluation Layer:** checking model behaviour using suitable metrics and validation.
- **Assessment Layer:** completing quizzes, assignments and the final assessment.

4.8 Methodology

The internship followed a progressive week-wise methodology. The main learning areas are summarised below.

Week	Module	Main Activity
1	Introduction to Data Science	Data science lifecycle, roles and Python introduction.
2	Python for Data Analysis	NumPy, Pandas, data manipulation and cleaning.
3	Data Visualization	Charts, plots and visual interpretation.
4	Statistics & Probability	Distributions, inference and hypothesis testing.
5	Machine Learning Basics	Supervised/unsupervised learning, regression and classification.
6	Advanced Machine Learning	Evaluation, feature engineering, ensembles and clustering.
7	Model Deployment & Tools	Deployment concepts and supporting tools.
8	Capstone Project	Applying the complete workflow to a real-world dataset.
Final	Credential Validation	Final assessment and internship grade evaluation.

4.9 Expected Outcomes

- Ability to use Python for practical data analysis.
- Working familiarity with data cleaning and transformation using Pandas and NumPy.
- Better understanding of data visualisation using Matplotlib and Seaborn.
- Improved familiarity with probability and statistical reasoning.
- Understanding of training and evaluation stages for machine learning models.
- Ability to connect data preparation, analysis and modelling into a single workflow.
- Successful completion of the internship, reflected by the Outstanding grade on the certificate.

4.10 Certificate of Completion

The supplied certificate confirms that **Anjali Kumari** successfully completed the eight-week Data Science Master Virtual Internship during April–June 2026. It records an **O (Outstanding)** grade, Certificate ID **420d992094f9d74a4558e**, and Student ID **anjali.kumari.cs28@iilm.edu**.

5. BIBLIOGRAPHY/REFERENCES

1. EduSkills Academy, Certificate of Virtual Internship – Data Science Master, 2026.
2. AICTE National Internship Portal.
3. EduSkills Foundation / EduSkills Academy.
4. Python Software Foundation, Python Documentation.
5. Pandas Documentation.
6. scikit-learn Documentation.